

Satellite Image and Received Signal-based Outdoor Localization using Deep Neural Networks

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Abstract—This paper proposes an outdoor localization scheme using image classification and deep learning by using information from satellite images and signals in the LTE band. We propose a two-step solution comprised of image differencing using satellite images to classify users into regions, and feed forward neural networks that take channel state information (CSI) data as an input to predict the user's longitude and latitude. We use real data collected from an LTE-A mobile communication system to assess the performance of our technique. Our results show an improvement in localization accuracy using the region based approach, despite the limited data provided from the single input, single output (SISO) system, as we achieve a mean absolute error (MAE) of 5.83 m. The proposed technique can be supplementary to GPS in urban canyons with weak GPS penetration.

Index Terms—Deep Neural Networks, localization, Received Signal Strength.

I. INTRODUCTION

With the push towards Internet of Things (IoT) applications and enhanced broadband services with reduced latency and higher bandwidth, localization has become a progressively important topic. The demand for localization based services is growing beyond only navigation based applications as emerging technologies such as AR/VR, social networking, crowd sourcing, online gaming and targeted advertising require the user's location [1]. Moreover, knowledge of the user's location is used in LTE and 5G networks to assist the base station with accurate estimation of channel conditions. This can also aid the base station by improving resource management and beam forming [2]. Although traditional localization techniques already exist, such as Global Positioning System (GPS) for outdoor localization and fingerprinting and proximity for indoor localization, these classical approaches can be costly and also come with their own set of challenges. GPS can achieve precise localization in outdoor line of sight (LOS) scenarios, however, its performance significantly degrades indoors and with the presence of any obstructions. This makes it less favourable to use in dense urban canyons, e.g. a downtown area with many tall buildings. Additionally, traditional fingerprinting approaches require precise calibration in the offline phase as well as many reference points to achieve good localization performance. These methods sometimes also suffer from the fluctuation of received signal strength (RSS) values as a result of multipath effects.

Recent studies have explored the use of machine learning-based solutions for routing, resource allocation and adaptive modulation [3] [4] [5]. Machine learning-based localization techniques have been proven to achieve state of the art

results, particularly in deep learning. The integration of these techniques with multiple input, multiple output (MIMO) systems has addressed some of the issues that were present in traditional localization approaches, such as the need for precise calibration in the offline phase and the fluctuation of instantaneous RSS values. Such techniques are discussed in [6] and [7]. In our prior work in [8], we proposed a localization technique for MIMO systems in 5G using LIDAR images and RSS data. In this paper, we extend the application of these techniques to single input, single output (SISO) systems by introducing a two-stage localization technique using image classification and deep learning for localization in an LTE SISO system. We divide the map of the localization area into 4 different regions. We then propose the use of image differencing using satellite images to classify users into one of the regions. The localization prediction is performed using feed forward neural networks that are distinct to each region. This approach allows the networks to learn the trends and outcomes specific to their region, despite the lack of data due to the single antennas in the SISO system. The neural networks are trained using channel state information (CSI) data, particularly signal to interference and noise ratio (SINR), reference signal received power (RSRP), reference signal received quality (RSRQ) and power over received signal strength indicator (RSSI).

This paper is organized as follows: section II presents the recent studies. Section III defines the proposed technique and the various adaptations of the algorithm in detail. Section IV starts by introducing the localization scenario and dataset used for the verification of the proposed methodology and also provides the achieved results. Finally, the paper is concluded in section V and future work is discussed.

II. RELATED WORK

While traditional outdoor localization techniques relies heavily on the use of GPS and indoor localization techniques explore the use of fingerprinting and proximity, and there has been a push towards machine learning based solutions in recent years. The need for more accurate localization methods is due to some of the shortcomings of the current technologies. For example, while GPS can achieve considerably low localization errors in outdoor unobstructed scenarios, its performance significantly degrades indoors and in dense urban areas where line of sight (LOS) transmission of signals is not feasible. On the other hand, fingerprinting methods suffer from the fluctuation of RSS values due to multi-path effects. These issues become

particularly imperative with the deployment of the up and coming 5G network as it utilizes the mmWave band whose attenuation properties cannot withstand non LOS transmission and significantly degrades with long distances.

Deep learning has been a popular topic in recent research and has proven to achieve state of the art localization accuracy while maintaining fairly low costs as the systems typically require fewer reference points than traditional fingerprinting techniques. The use of Neural networks, Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNN) is explored in [9], [6] and [7]. [6] and [7] also provide a comparison between deep learning approaches and classic machine learning techniques like linear regression (LR) and K-Nearest Neighbours (KNN).

[9] proposes an outdoor localization scheme using deep learning from data collected using Long-Range Wide-Area Network (LoRaWAN) communication protocol. The authors aim to improve fingerprinting-based localization in complex indoor and outdoor environments.

[6] adopts a neural network based approach in a massive Multiple Input, Multiple Output (MIMO) system to predict locations, the performance of the deep learning approach is then compared to a more classical KNN learning algorithm.

[7] also takes a similar approach in which they implement various supervised machine learning techniques such as linear regression (LR), weighted K-nearest neighbours (WKNN) and neural networks (NN) to estimate users locations outdoors by observing signals transmitted by single antenna users to a massive MIMO system.

Our proposed technique differs from the above studies in that it extends the application of deep learning for localization to a SISO system with only 4 features per sample. We also propose a vision-aided solution that takes advantage of satellite images displaying the receiver's location to classify the users into a general region in the map prior to performing precise localization. This approach helps the neural networks training process despite the lack of data and features available, as they are trained with samples that are distinct to their regions.

III. PROPOSED DNN BASED LOCALIZATION TECHNIQUE

We propose a two-stage localization scheme that first predicts the user's general location using the provided satellite images, then using a regional FFNN, predicts their longitude and latitude. The regional classification is performed using image differencing, an algorithm typically used to detect change between satellite images. The FFNN takes the matrix described in eq. 1 as an input, and provides the normalized longitude and latitude as an output. The longitude and latitude are then extracted from the network's output using reverse normalization. The regional classification and FFNN-based localization processes are described in Fig. 1 and algorithm 1. Satellite training images are used by the image differencing algorithm, described in algorithm 1, to compute the difference between the images and classify the region. The regional classification and FFNN based localization stages are described in greater detail below.

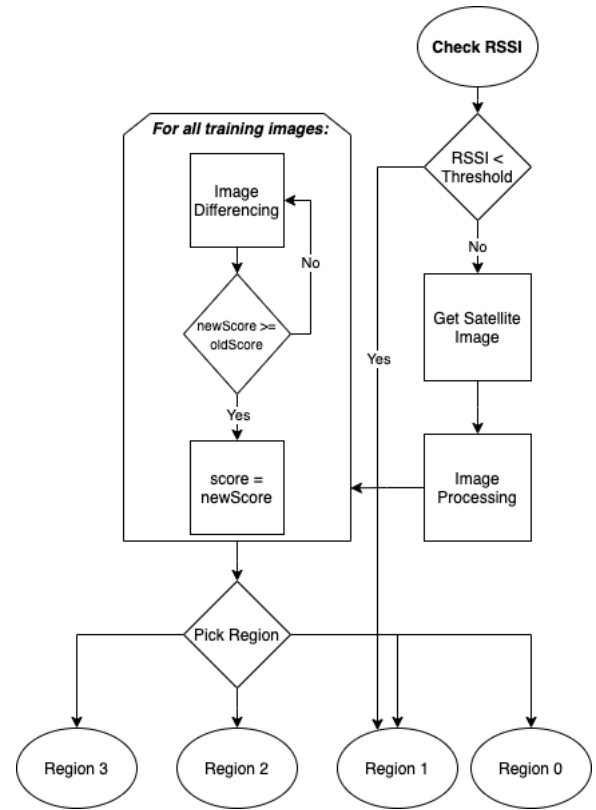


Fig. 1. Flowchart describing regional classification scheme

A. Regional Classification

The regional classification stage uses image differencing to allocate users into a general, pre-defined region in the map. Image differencing is a technique used to detect changes between images, that works by comparing the images pixel by pixel and calculating the squared error of the difference between the original and test image [10]. The algorithm uses the Structural Similarity Index Matrix (SSIM) which outputs a score that characterizes the resemblance between two images; the closer the score is to 1, the stronger the resemblance. We split the map of the localization area into 4 different regions, shown in fig. 2.

Our proposed image differencing algorithm takes the processed satellite image as an input and compares it to a stored repository of training images, each representing a different region. The SSIM score between the input image and each training image is calculated. The input image then belongs to the region of the training images that yields the highest score. Several images from varying parts of the region are used as training samples for the image training repository. 65% of the training images from each region were randomly sampled and used to represent the landmarks in that region. We also propose a second adaptation of the algorithm that uses a hybrid regional classification approach that integrates the image differencing algorithm with RSSI-based localization. To implement this, we started by plotting the RSSI strength over the localization area, shown in fig. 4. This plot showed

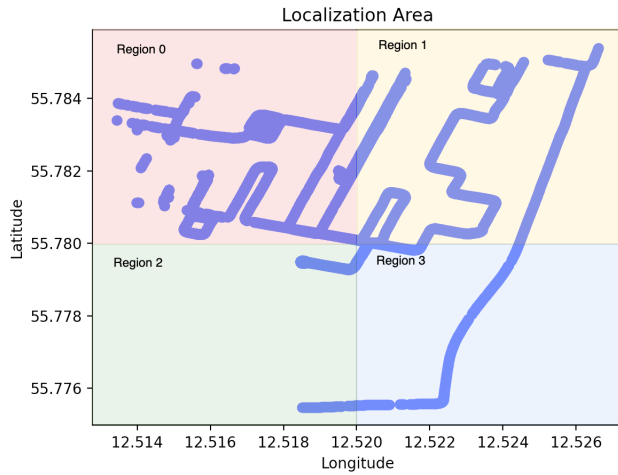


Fig. 2. Image displaying the solid regional boundary approach for the proposed localization scheme

that the RSSI values are strongest in region 1. Using this information we created a simple comparison algorithm that checks the sample's RSSI; if the RSSI is greater than a pre-defined threshold then the sample is sorted into region 1, otherwise the image differencing technique is used. The RSSI threshold is set based on trial and error measurements that produce the greatest localization accuracy.

We also consider two approaches to defining the regional boundaries: Regions with solid boundaries as previously shown in fig. 2 and regions with overlapping boundaries. The overlapping boundaries approach extends the region's boundaries by approximately 50 m, creating an overlap between neighbouring regions. This is intended to decrease the amount of false classifications that can occur at regional boundaries due to the similarity of landmarks.

Algorithm 1 Image Differencing Algorithm

```

Data: Satellite Images
Result: Regional Label
initialization for Every Input Image do
    for Every Training Image do
        checkScore(currentImage, TrainingImage)
        score = currentScore
        if newScore  $\geq$  currentScore then
            score = newScore
        end
    end
    Region = index(max(imageScore))
end

```

B. FFNN Based Localization

In our scheme, FFNNs are used to predict the location of users. The dataset uses a SISO system, hence there are only measurements from a single antenna. The lack of data can cause a more complex neural network structure like CNNs to overfit. We propose the use of 4 different FFNNs, one for

each region of the map. The architecture of the networks is summarized in table I. Each FFNN takes the radio parameters data shown in eq. 1 as an input and is trained with data samples that are specific to that region. Dropouts were added to the networks of the regions that were over-fitting. Dropout regularization works by randomly ignoring or dropping neurons in layers. They prevent network layers from co-adapting to prevent mistakes from previous layers, hence producing sturdier results [11]. The proposed network architecture also uses Adam optimizer for the back-propagation algorithm. Adam is a method for stochastic optimization that replaces stochastic gradient descent, as it combines the advantages of adaptive gradient algorithm and root mean square propagation [12]. The implementation of all the networks is done in python using the Tensorflow library.

TABLE I
TABLE SUMMARIZING THE NETWORK ARCHITECTURE OF THE FFNN FOR THE LOCALIZATION SCHEME

Parameters	Values			
	0	1	2	3
Architecture	128x64x 32x16x3	128x64x 32x16x3	128x64x 32x16x3	128x64x 32x16x3
Epochs	5000	3000	6000	5000
Batch Size	10	10	10	10
Dropout	0.2	N/A	0.2	N/A
Activation Function Hidden Layers	Sigmoid	Sigmoid	Sigmoid	Sigmoid
Activation Function Output Layer	Linear	Linear	Linear	Linear
Loss Function	MSE	MSE	MSE	MSE

IV. PERFORMANCE EVALUATION

A. Dataset

The dataset provides real measurements from an LTE-A communications system that is located at the University of Denmark [13]. The scenario consists of one mobile vehicle with a single antenna and one fixed base station with 3 PCIs, 2 at 811 MHz (PCI_64 and PCI_65) and 1 at 2630 MHz (PCI_302). The overall coverage area is approximately 2.4 Km by 1.25 Km, the map of the localization area is shown in Fig. 3. There are 57585 samples in total, however, the overall dataset is split into 3 different sets based on the PCI index. For the purpose of this study, only PCI_64 is used which contains 11863 samples and has a coverage area of 1.446 Km by 1 Km shown in fig. 3. The dataset contains SINR, RSRP, RSRQ, Power/RSSI and a satellite image for each sample.

The pre-processing stage of the channel state information consists of min max normalization and the concatenation of the data into a single vector. This is a SISO system, hence a one dimensional vector is used in place of the image feature matrix. The structure of the input vector is shown in Eq.1.

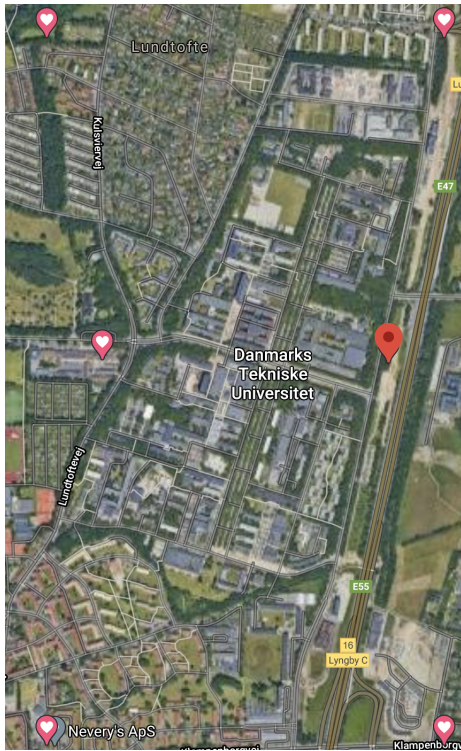


Fig. 3. The area of localization data collected at University of Denmark,

$$[SINR \ RSRP \ RSRQ \ Power/RSSI] \quad (1)$$

The provided satellite images are used for the image classification stage. All of the images have the same dimensions of 256 x 256 pixels. The preprocessing stage consists of converting the images to gray scale to reduce dimensions, blurring the images to remove sharp edges and details, converting the images to black and white based on a defined colour threshold and finally, resizing the images to reduce computational complexity. The process is shown in Fig. 5.

We plotted the RSSI values to analyze the correlation between location and RSS strength. It is determined that region 2 contains the strongest RSSI values in these datasets. Regions 1, 3 and 4 have weaker RSSI values, making them more difficult to classify using fingerprinting. This is shown in fig. 4. This information is later used in the proposed mechanism to identify the areas that can be classified into regions using the RSSI value, rather than the satellite images.

B. Results

This section assesses the performance of the proposed methodology in section III by applying the algorithms to the dataset. To quantify the performance, the following metrics are used:

- **Localization Accuracy:** The ability of the algorithm to accurately predict locations. In this paper, the localization accuracy is calculated using root mean square error (RMSE) and mean absolute error (MAE), shown in eq. 2 and 3, respectively.

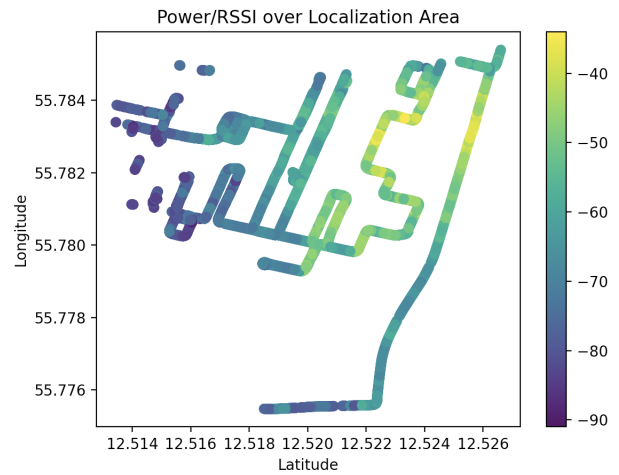


Fig. 4. Power/RSSI measurements over the localization area

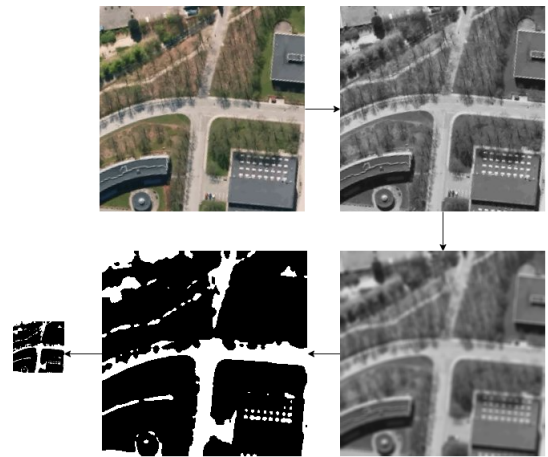


Fig. 5. Image pre-processing stages for the proposed localization scheme

- **Latency/ Delay:** The system should be able to locate users without any noticeable delay, ideally in milliseconds time-frame, in order to be feasible in a real-time localization scenario [14].
- **Availability:** Use of hardware that is readily available on users devices, rather than creating a new infrastructure [14].
- **Cost:** The cost of the localization system is also affected by the need for new hardware. Similar to the availability metric, the system should not require a new infrastructure or costly devices.
- **Energy Efficiency:** The systems ability to minimize energy consumption and not drain user's devices. Several factors can affect energy efficiency such as frequency of transmission required for localization, transmission power and computational complexity [14].
- **Reception Range:** The system's ability to overcome signal degradation and interference as a result of a larger range.
- **Scalability:** The system should be easily adaptable to new

environments, and scalable to a large number of users.

In this paper, RMSE and MAE are used as the error metric to measure localization accuracy, the equations are described below.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y - x)^2}{n}} \quad (2)$$

$$MAE = \frac{\sum_{i=1}^n |y - x|}{n} \quad (3)$$

We assess the performance of the algorithm with different adaptations of the algorithm: image classification with solid boundaries, image classification with overlapping boundaries, hybrid classification with solid boundaries and hybrid classification with overlapping boundaries. The hybrid classification approach is the adaptation of the algorithm described in section III, which integrates the use of both image differencing and RSS based fingerprinting to classify the data samples into their respective regions. The singular FFNN is used as a benchmark to showcase the improvement in localization error as a result of the regional approach. As this dataset provides the user positions in longitude and latitude format, Eq. 4 is used to calculate the distance between two points. The RMSE and MAE are then calculated using Eqs. 2 and 3 respectively.

$$\begin{aligned} a &= \sin^2(\Delta\psi/2) + \cos\psi_1 \cdot \cos\psi_2 \cdot \sin^2(\Delta/2) \\ c &= 2 \cdot \operatorname{atan2}(\sqrt{a}, \operatorname{sqrt{1-a}}) \\ d &= R \cdot c \end{aligned} \quad (4)$$

Where R is the Earth's radius
 $R = 6371 \text{ Km}$

The classification accuracy demonstrates the successful attempt at performing regional classification of the data samples using the satellite images. The algorithm achieves an accuracy of 99.37%, 99.40%, 99.52% and 99.97% for the hybrid regional classification with solid boundaries, image classification with solid boundaries, hybrid regional classification with overlapping boundaries and image classification with overlapping boundaries approaches, respectively. The results show that the different adaptations of the algorithm have negligible effects on the classification accuracy. This is likely due to the amount of detail in the satellite images which makes the landmarks easier to distinguish. However, the introduction of the hybrid regional classification also produced insignificant improvements to the classification accuracy. As such, the main benefit of this approach remains to be the reduced computational complexity.

The results of each adaptation, for every region, are summarized in Table II. We use both RMSE and MAE as metrics to assess the performance of the algorithm to gain more insight. The RMSE applies more weight to large errors, thus allows us to know when there are significant variations between the predicted and actual locations. The MAE describes the average performance. In Table II, a common trend between all adaptations as well as regions is the higher RMSE value

with respect to MAE. Various factors can contribute to this. For example, false regional classifications can result in the sample being trained by the wrong neural network, hence producing an inaccurate prediction and causing large errors. Moreover, as the proposed methodology is being applied to a SISO system, the lack of input data to the neural network limits its learning capabilities. However, an improvement in the system's learning capabilities can be seen through the use of the regional FFNNs. The results show a significant improvement in MAE and RMSE with the use of the regional FFNN approach in comparison to the singular FFNN as the singular FFNN achieves an MAE of 15.74 m, whereas the regional FFNN achieves a minimum MAE of 5.83 m.

In summary, the proposed technique achieves promising localization accuracy that is comparable to recent studies. The improvement in accuracy as a result of the region based FFNN approach is demonstrated in Table II as the singular FFNN achieves an MAE of 15.74 m and the regional approach achieves a MAE of 5.83 m. The performance of the proposed methodology is also assessed against the metrics previously described. This is discussed in more detail below:

- **Latency/ Delay:** Due to the larger and more detailed satellite images, the image differencing algorithm achieves a processing and prediction time of 26.7 s. The results display a trade-off between regional classification accuracy and latency. The judgement between classification accuracy and latency can be made depending on the application of the system and the speed of receiver. For example, a system with mobile receivers is less susceptible to longer delays whereas, a vehicular communications system with fast moving cars would require very-low latency for successful localization.
- **Availability:** The proposed system takes advantage of current LTE and satellite infrastructure.
- **Cost:** There are no additional costs associated with the implementation of this localization system due to the abundant availability of communication systems in most vehicles and devices today. Additionally, the proposed methodology also uses satellite images from the pre-existing satellite infrastructure.
- **Energy Efficiency:** The proposed system takes advantage of information that is readily available to the base station by forming a communication link, such as power/RSSI, RSRP, RSRQ and SINR. Hence, the number of transmissions for localization can be reduced. The use of machine learning eliminates the need for complex computations that are traditionally used in localization techniques. Although the networks need to be trained, which can take up to a few minutes, this process only needs to be completed once. Moreover, the localization algorithm takes place at the base station rather than the user's devices, which reduces impact to their battery life.
- **Reception Range:** The system is capable of overcoming signal degradations and environmental obstructions as the provided dataset take place in outdoor scenarios. More-

TABLE II
TABLE SUMMARIZING RESULTS OF THE LOCALIZATION SCHEME FOR EACH ADAPTATION

	Hybrid Classification w/ Solid Boundaries		Image Classification w/ Solid Boundaries		Hybrid Classification w/ Overlapping Boundaries		Image Classification w/ Overlapping Boundaries		Singular FFNN	
	RMSE (m)	MAE (m)	RMSE (m)	MAE (m)	RMSE (m)	MAE (m)	RMSE (m)	MAE (m)	RMSE (m)	MAE (m)
Region 0	25.77	4.533	8.51	2.96	12.99	6.77	10.62	4.94	NA	NA
Region 1	19.48	10.74	19.55	8.88	18.79	5.98	18.87	8.07	NA	NA
Region 2	23.4	9.33	5.61	3.65	28.67	9.77	7.55	5.16	NA	NA
Region 3	24.38	12.94	11.01	8.24	20.56	10.72	20.18	10.34	NA	NA
Total	16.31	9.25	14.29	5.83	15.24	8.41	14.94	6.55	32.36	15.74

over, The proposed system uses a centralized method with only one base station, without the need for any excess reference points or repeaters. The dataset covers a terrain size of 1.446 Km by 1 Km.

- **Scalability:** The scalability of the algorithm is demonstrated through its application to a dataset of a size of at least 10,000 samples. The performance of the proposed methodology was applied to a map at the University of Denmark using real data from a SISO LTE system, demonstrating its feasibility in a real-life system. Although the trained neural networks rely heavily on the datasets, the system can be easily updated with new data as the radio environment evolves.

V. CONCLUSION

This paper introduced a localization technique using a two-stage solution comprised of regional classification and deep learning using FFNNs and satellite images. Two interpretations of regional classification are proposed: a hybrid approach using fingerprinting and image differencing and standalone image differencing. The proposed methodology also explored the use of two ways of defining regional boundaries: regions with solid boundaries and regions with overlapping boundaries. The algorithm is applied to the Mobile Communication System and Satellite Images dataset [13] which contains real measurements from an LTE-A SISO system at the University of Denmark. The University of Denmark's map is split into 4 regions. We use the satellite images to classify the user's region and the FFNNs to predict the location co-ordinates. Satellite images are used to classify the user's region and the FFNNs is used to predict the co-ordinates. The results showed that the use of the region based FFNNs produced significant improvements in localization accuracy in comparison to the singular FFNN. Overall, the proposed methodology shows promising results. Although the use of the SISO system limits the number of features that the neural networks are able to learn from, the proposed methodology achieved a localization error of 5.83 m, which is comparable to most of the related studies using MIMO systems and even outperforms some.

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